

Parking Management System for Urban Parking Space

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Abstract

This document addresses the growing challenge of urban parking management, with a particular focus on the underutilization of parking spaces at the FH Dortmund SON campus during off-peak hours. The central inquiry guiding this study is as follows: What technological solution can be implemented to effectively manage the parking spaces at the FH Dortmund SON campus during non-working hours for the benefit of both employees and local residents? The study proposes a smart parking system that leverages the Internet of Things IoT technology, integrating existing Bosch Parking Sensor PLS110 V2.1 sensors and Elka ES40 gates with a web-based application to optimize parking space utilization.

The proposed methodology involves the development of a web application using the React framework that allows users to register, reserve, and pay for parking spaces in real time. The system employs a dynamic pricing model to optimize revenue generation and ensure equitable access, while facilitating secure entry through QR code authentication and the integration of IoT-enabled gates. The implementation also includes real-time availability tracking, user management, and secure payment processing.

In conclusion, this work provides a valuable technological solution to urban parking challenges by improving the efficiency of existing infrastructure. The system not only addresses the immediate parking management needs of the FH Dortmund SON campus but also provides a scalable model that can be adapted to other urban environments facing similar issues. The integration of the Internet of Things (IoT), real-time data processing, and user-friendly interfaces represents a significant advancement in smart parking systems, contributing to sustainable urban development.

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Abbreviations

FH	Fachhochschule Dortmund
SON	Sonnenstraße
IoT	Internet of Things
EV	Electric Vehicle
TCP/IP	Transmission Control Protocol / Internet Protocol
QR	Quick Response
MQTT	Message Queuing Telemetry Transport
GPIO	General Purpose Input / Output
UART	Universal Asynchronous Receiver-Transmitter
API	Application Programming Interface
LED	Light Emitting Diode
JSON	JavaScript Object Notation
NGSI	Next Generation Service Interface

1 Introduction

The number of automobiles in urban areas has increased exponentially over time. [1] This has resulted in several issues, including a shortage of parking spaces. At the SON campus of Dortmund University of Applied Sciences and Arts, there are parking lots designated for university staff; however, these entrances are gated, and access is restricted to individuals with personal badges. During non-working hours and on weekends, the university parking lots are sparsely occupied, and in some cases, they are completely unoccupied. However, these lots are inaccessible to individuals residing near the university. This document will explore a technological solution to this problem.

1.1 Research Question

What technological solution can be implemented to effectively manage the parking spaces at the FH Dortmund SON campus during non-working hours to benefit both employees and local residents?

2 Research and Methodology

A substantial challenge confronting contemporary urban centers pertains to the proliferation of vehicles, which, in turn, engenders an augmented demand for parking spaces. Germany has been grappling with this issue for decades, as evidenced by the following statement: *"The primary cause of this predicament is the substantial increase in motorization over the past few decades. The existing supply of parking space has proven inadequate in meeting the increasing demand for parking."* [2] In order to address this challenge, it is necessary to utilize the unused parking spaces on the SON campus during off-hours. This objective can be accomplished by implementing an Internet of Things (IoT)-based system, which has the potential to facilitate efficient and user-friendly parking management for local residents. The proposed system would integrate the existing Fireware [3] infrastructure database of FH Dortmund with a web-based software platform. This platform will enable users to register, log in, and access designated parking spaces. The FH Dortmund is equipped with ten Bosch Parking Sensors (PLS110 V2.1), which will be installed in the parking slot areas to monitor their availability during off-hours. Real-time tracking will be confined to these monitored spaces to ensure efficient utilization. Other parking areas, while not subject to real-time monitoring, will remain accessible to users as long as they are unoccupied.

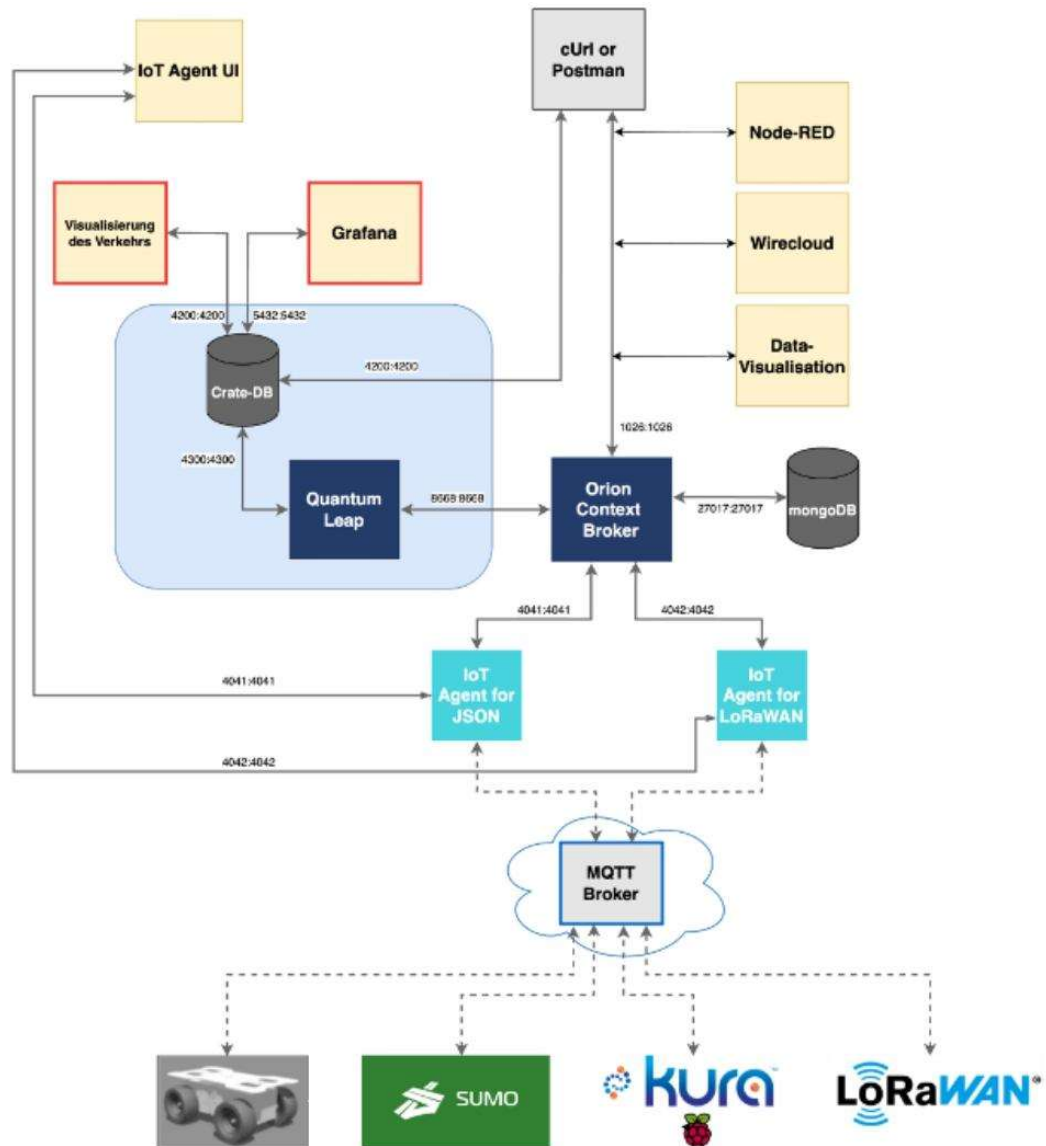


Illustration 1: FH Dortmund FIREWARE System.

Source: Salim Sir (2025)

2.1 Smart Parking System

A smart parking system employs Internet of Things (IoT) technology to address inefficiencies in parking space utilization by integrating sensors, databases, and software platforms. For the SON campus, the proposed system will utilize the existing Bosch Parking Sensors to monitor parking space availability in real time. These sensors will transmit data to a central database connected to FH Dortmund's FIREWARE infrastructure, allowing seamless integration with the proposed web application.

To ensure controlled access, the parking lots are equipped with Elka ES40 gates with MO64 controllers. [4] [5] In order to integrate these gates into the proposed

system, it is necessary to install an IoT module, which will enable remote opening through the application. This enhancement will allow users to enter the parking area efficiently, ensuring that only those with valid reservations can access the facility. This configuration facilitates precise tracking of parking spaces, reduces the time spent searching for available spots, and minimizes unauthorized access by restricting use to registered users. The system's automation capabilities will enhance the overall user experience and operational efficiency.

2.2 Dynamic Pricing Model

Parking facilities on the SON campus will be available to local residents during non-working hours, specifically from 6:00 p.m. to 7:00 a.m. on weekdays and throughout the weekend. This scheduling approach is designed to ensure that the parking facilities remain predominantly accessible to university employees during standard office hours, while concurrently maximizing their utility during off-peak periods. By opening the lots during these hours, the system addresses the pressing need for residential parking without interfering with the University's operational needs. This arrangement is consistent with the dynamic pricing model, as it utilizes incentives to regulate usage during the designated hours.

The implementation of a dynamic pricing model is intended to optimize parking utilization and balance demand. The parking rates will be subject to variation on the basis of day of the week and time of day, with higher rates during periods of peak demand and lower rates during off-peak periods. Proactive pricing, which employs predictive analytics to inform pricing decisions, has been shown to yield superior outcomes due to its capacity for optimization. [6] The following assertion is made:

Regarding standard parking spaces, the recommended rate will be €1.50 per hour. This rate is consistent with typical urban parking rates in Germany, which range from €1 to €3 per hour, depending on the city and demand. The implementation of a moderate rate is a strategy that ensures accessibility while effectively managing demand.

In the case of disabled parking spaces, the implementation of a moderate rate will result in their being free of charge. In Germany, individuals with disabilities who possess a valid EU parking card are legally entitled to unrestricted and complimentary parking in designated areas, provided that no alternative parking

options are available in the immediate vicinity. This policy is consistent with the principles of inclusivity and is in accordance with national regulations.

The proposed rate for electric vehicle (EV) charging points is €2.50 per hour. It is important to note that EV charging stations frequently exhibit elevated operating costs, attributable to significant electricity consumption and the necessity of maintenance. A slightly higher rate is employed to reflect these costs while encouraging the adoption of electric vehicles. Failure to comply with designated parking regulations for electric vehicles (EVs) can result in penalties, including a fine of €55, thereby underscoring the necessity for adherence to these guidelines.

2.3 Web Application

The user interface and functionality of the parking system will be delivered through a web application developed using React, a modern web framework optimized for performance. The application will feature a user-friendly design, enabling residents and university staff to easily register, log in, and reserve parking spaces. Key functionalities of the app include:

Constant availability display: Users can view the availability of monitored parking spaces.

Dynamic pricing integration: The app will display current parking rates based on the dynamic pricing model.

User management: The app will maintain a database of registered users, enabling seamless access control and monitoring of the usage of the parking lot.

Availability prediction: The users can view the prediction of the parking lot usage for all the week, this allows them to know the estimated availability of parking spaces.

3 Implementation

3.1 Booking Application

The application will have 4 different sections, namely Login/Registration, Reservations, QR Profile and Payments.

The Login section allows you to log in or register with the application. This action is required to make and manage parking reservations.

3.1.1 Reservations

The reservation system allows users to view available parking spaces in real time using the Bosch Parking Sensors that will be integrated into 10 different parking spaces and using Orion Context Broker [7] will update the reservations status. To park a vehicle during the designated hours, a reservation must be made through the application; otherwise, access to the parking lot will be denied, ensuring that reserved spaces are respected.

Upon selection of a parking space, the system temporarily reserves the space for a period of five minutes to prevent other users from making reservations. If the user does not confirm the selection within the designated time frame, the space is automatically released back into the system.

Once a reservation is confirmed, it is stored in the user's profile. It is important to note that each user is limited to maintaining only a single active reservation at any given moment. The reserved space remains reserved for a period of up to 10 minutes after the designated reservation time. If the user does not arrive at the designated parking location within this time frame, the reservation will be automatically canceled, and the space will become available for use by other users. This system is designed to ensure equitable access to parking spaces and to prevent the prolonged reservation of spaces that are not subsequently used.

This section also displays predictions of parking usage up to one week from the current day. By furnishing an estimated forecast of future availability, users can make informed decisions regarding their parking needs and more effectively plan their visits.

3.1.2 QR Profile

The QR Profile section of the application displays a QR code on the user's screen that must be scanned at the parking lot entrance gate using a Fingertec QR 110

[8] reader (specified on section 3.4). This QR code contains the logged-in user's information, which is checked against the database. If the user has a valid reservation, access to the parking lot is granted.

This system provides secure and efficient access control, preventing unauthorized vehicles from entering and ensuring that only users with confirmed reservations can use the parking spaces during the designated hours.

3.1.3 Payments

The payment section of the application provides a secure and convenient payment gateway, allowing users to select their preferred payment method to complete their transactions. The system supports multiple payment options, including credit and debit cards, PayPal, and digital wallets such as Google Pay and Apple Pay.

To ensure a seamless user experience, the platform will integrate a secure authentication system that protects users' financial information through encrypted transactions. In addition, users will be able to save their payment details for faster future transactions. The system will generate digital receipts for each completed payment, which can be accessed from the user's profile.

This flexible and secure payment system ensures accessibility to a wide range of users while maintaining high standards of security and convenience in the parking management process.

3.2 IoT Gate Integration

The ELKA ES40 barrier constitutes an electromechanical gate control system that has been developed for the purpose of managing parking facilities. Given the university's existing private parking system, access to the facility is controlled by gates that require a unique and personal badge for entry. To integrate the doors with the application, an IoT module must be installed to connect to the Internet and open the doors when a user wants to use the service. Two solutions were devised to integrate the barrier with an IoT infrastructure: the utilization of the controller that is already integrated in the barrier (MO64). The MO64 controller possesses the capacity to function through relay-based actuation, wherein an electrical signal initiates the movement of the barrier. It is equipped with TCP/IP connectivity,

facilitating remote control via a network or the internet, and RS485 communication, a wired interface that enables long-distance data exchange with other devices.

3.2.1 ELKA TCP/IP Module for MO64

The ELKA TCP/IP [9] is a module that facilitates remote access to the barrier system by enabling network communication through standard TCP/IP protocols. This integration enables external control of the gate through the application and circumvents the necessity for extensive integration with the FH's FIREWARE infrastructure.

3.2.2 ESP32 Microcontroller

The ESP32 [10] This microcontroller boasts a high degree of versatility, featuring built-in Wi-Fi and Bluetooth connectivity, positioning it as a promising candidate for integrating IoT functionality. Its programmable capabilities allow for seamless interfacing with the MO64 controller via GPIO (for relay-based control), UART (for serial communication), or HTTP/MQTT (for direct cloud-based interactions), facilitating seamless integration with the FH's FIREWARE System.

3.3 Gate Opening

The React-based application serves as the user interface for managing parking operations. It interacts with the Fireware system and the barrier control components through the following mechanisms:

1. User Authentication & QR Code Validation:
2. Users generate a QR code in the React application.
3. A QR scanner at the parking entrance reads the QR code and sends the data to the FIWARE system.
4. The Orion Context Broker verifies the validity of the QR code and triggers the ESP32 or ELKA TCP/IP module to open the barrier.

Manual Override & Remote Access:

- Administrators can manually trigger the barrier from the React application by sending an API request to FIWARE, which then relays the command to ESP32 or ELKA TCP/IP module.

3.4 QR Reader

The Fingertec QR 110 is a compact QR code reader designed specifically for access control applications, while the Ingressus [11] Ethernet module acts as an intermediary, facilitating communication between access control devices and centralized authentication systems.

To integrate QR 110 and Ingressus into the existing FIWARE architecture, the following is required:

1. The QR 110 reader captures QR codes from users mobile applications, while Ingressus processes and transmits this data for authentication.
2. An IoT agent-such as one that supports JSON or MQTT can be used to convert the QR scan data to NGSI format before passing it to the Orion Context Broker.
3. The Orion Context Broker, in turn, manages authentication requests by matching the scanned QR codes against a pre-existing database, potentially in MongoDB or an alternative identity verification system.

3.5 Availability Signaling

The integration of a WS2812B addressable LED strip with an ESP32 microcontroller provides a highly efficient and scalable solution for real-time parking status indication, it will be installed in all parking slots that have a Parking Sensor installed. This system uses MQTT to enable seamless communication between the ESP32 module and the existing FIWARE infrastructure.

Upon successful reservation through the application, the status of the corresponding parking space is updated accordingly. This mechanism serves to prevent potential conflicts between users with physical access credentials and those using the application to make reservations. It also provides a clear visual indication of which spaces are currently reserved, and which are available. To further improve clarity and organization, each parking space has a unique number painted on the floor, ensuring that users can easily locate their assigned space.

4 Conclusions

This document presents a comprehensive solution to the growing challenge of urban parking management, specifically addressing the underutilization of parking spaces at the FH Dortmund SON campus during non-working hours. The proposed smart parking system leverages IoT technology by integrating existing Bosch Parking Sensors and Elka ES40 gates with a web-based application developed using the React framework. The system aims to optimize parking space utilization, improve the user experience and generate additional revenue through a dynamic pricing model.

The implementation of the system includes real-time availability tracking, user management, secure payment processing, and controlled access through QR code authentication. The dynamic pricing model, which varies rates based on demand and time of day, ensures equitable access and incentivizes off-peak parking usage. The web application provides an easy-to-use interface for registering, reserving, and paying for parking, while also offering features such as real-time availability and parking usage forecasts.

The integration of IoT-enabled gate control systems and QR code-based authentication ensures secure and efficient access to parking facilities, preventing unauthorized use and minimizing conflicts between users. The system's scalability and adaptability make it a valuable model for other urban environments facing similar parking challenges.

In summary, this work contributes to sustainable urban development by improving the efficiency of the existing parking infrastructure. The proposed system not only addresses the immediate parking management needs of the SON campus but also provides a scalable and innovative solution that can be adapted to other urban environments. The integration of IoT, real-time data processing and user-friendly interfaces represents a significant advancement in smart parking systems, providing a practical and effective approach to urban parking management.

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